PRANVEER SINGH INSTITUTE OF TECHNOLOGY, KANPUR

DEPARTMENT OF COMPUTER SCEINCE & ENGINEERING

Odd Semester 2025-26

B. Tech.- Fourth Year

Semester - VII

Lab File

Artificial Intelligence Lab

(BCS 751)

Submitted To:

Faculty Name :________________

Designation :________________

Submitted By:

Name :________________

Roll No. :________________

Section :________________

Table of Contents

Department Vision Statement

To be recognized, the Department of Information Technology produces versatile computer engineers, capable of adapting to the changing needs of computer and related industry.

Department Mission Statements

The mission of the Department of Information Technology is:

i. To provide broad-based education with knowledge and attitude to succeed in Information Technology careers.

ii. To prepare students for emerging trends in computer and related industry.

iii. To develop competence in students by providing them with skills and aptitude to foster culture of continuous and lifelong learning.

iv. To develop practicing engineers who investigate research, design, and find workable solutions to complex engineering problems with awareness & concern for society as well as the environment.

Program Educational Objectives (PEOs)

i. The graduates will be efficient leading professionals with knowledge of Information Technology discipline that enables them to pursue higher education and/or successful careers in various domains.

ii. Graduates will possess the capability of designing successful innovative solutions to real life problems that are technically sound, economically viable and socially acceptable.

iii. Graduates will be competent team leaders, effective communicators and capable of working in multidisciplinary teams following ethical values.

iv. The graduates will be capable of adapting to new technologies/tools and constantly upgrading their knowledge and skills with an attitude for lifelong learning

Department Program Outcomes (POs)

The students of the Information Technology department will be able:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, Information Technology fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and Information Technology sciences.

3. **Design/development of solutions:** Design solutions for complex Information Technology problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.

4. **Investigation:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex Information Technology activities with an understanding of the limitations.

6. **Engineering and Society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice in the field of Information Technology.

7. **Environment and sustainability:** Understand the impact of the professional Information Technology solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Information Technology practice.

9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Communication:** Communicate effectively on complex Information Technology activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, making effective presentations, and giving and receiving clear instructions.

11. **Project management and finance:** Demonstrate knowledge and understanding of the Information Technology and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. **Life-long learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Department Program Specific Outcomes (PSOs)

The students will be able to:

1. Use algorithms, data structures/management, software design, concepts of programming languages and computer organization and architecture.

2. Understand the processes that support the delivery and management of information systems within a specific application environment.

Course Outcomes

*Level of Bloom's Taxonomy	Level to be met	*Level of Bloom's Taxonomy	Level to be met
L1: Remember	1	L2: Understand	2
L3: Apply	3	L4: Analyze	4
L5: Evaluate	5	L6: Create	6

CO Number	Course Outcomes
BCS-751.1	Define **[1. Knowledge]** and apply basic search algorithms using Python.
BCS-751.2	Discuss **[2. Comprehension]** the scope of Artificial Intelligence to students where they design and implement logic-based knowledge representation and reasoning programs using Python and Prolog
BCS-751.3	Apply [**3. Application]** Demonstrate AI techniques for real-world problems such as chatbot development, planning, constraint satisfaction & probabilistic reasoning.

List of Experiments

Sr.No.	Objectives	CO
1	Implement Breadth First Search (BFS) for a given graph or maze.	CO1
2	Implement Depth First Search (DFS) for a tree or graph structure.	CO1
3	Solve the 8-Puzzle Problem using A* Search Algorithm.	CO1
4	Implement Hill Climbing Algorithm for numerical optimization or pathfinding.	CO1
5	Implement Simulated Annealing Algorithm for constraint-based search problems.	CO1
6	Solve Water Jug Problem using state-space search (BFS or DFS).	CO1
7	Write Prolog programs to define family relationships using predicates.	CO2
8	Implement 4-Queens Problem in Prolog using backtracking.	CO2
9	Implement Unification Algorithm in Python or Prolog.	CO2
10	Implement Forward and Backward Chaining in a rule-based system (manual or code-based).	CO2
11	Demonstrate Resolution in Propositional Logic through a basic example (e.g., proving a theorem).	CO2
12	Perform stemming and lemmatization on user-input text.	CO3
13	Implement Tic-Tac-Toe game with an AI opponent.	CO4
14	Implement Minimax Algorithm for turn-based games.	CO3
15	Enhance the game with Alpha-Beta Pruning.	CO3

INDEX

Sr.No.	Experiments	Date
1	**Implement Breadth First Search (BFS) for a given graph or maze.**	
2	**Implement Depth First Search (DFS) for a tree or graph structure.**	
3	**Solve the 8-Puzzle Problem using A* Search Algorithm.**	
4	**Implement Hill Climbing Algorithm for numerical optimization or pathfinding.**	
5	**Implement Simulated Annealing Algorithm for constraint-based search problems.**	
6	**Solve Water Jug Problem using state-space search (BFS or DFS).**	
7	**Write Prolog programs to define family relationships using predicates.**	
8	**Implement 4-Queens Problem in Prolog using backtracking.**	
9	**Implement Unification Algorithm in Python or Prolog.**	
10	**Implement Forward and Backward Chaining in a rule-based system (manual or code-based).**	
11	**Demonstrate Resolution in Propositional Logic through a basic example (e.g., proving a theorem).**	
12	**Perform stemming and lemmatization on user-input text.**	
13	**Implement Tic-Tac-Toe game with an AI opponent.**	
14	**Implement Minimax Algorithm for turn-based games.**	
15	**Enhance the game with Alpha-Beta Pruning.**	